Subsample_Incidence

Builds the epidemiologically-informed replica datasets used to test whether sampling bias between the two geographic locations (roughly 3:1 CDMX vs MEX) affects the results.

Each replica keeps all background sequences plus a randomly down-sampled fraction of the CIENI-CEC sequences (n = 1,896). The number kept per year per location is set by the lowest sequences-to-incidence ratio across the year × location cells, where incidence is new HIV cases per 100,000 inhabitants per year as officially reported by DGE, independently of CIENI-CEC. The per-cell targets are in Supplementary File 1, Supplementary Table 1.

Associated with the preprint

Proposed title: "HIV Transmission Dynamics Reveal Extensive Spatial Mixing Across Greater Mexico City"

Posted as:

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Files

File	What it is
`HIV_subsample_incidence.py`	Reads `input.fasta`, keeps sequences from `ST_1` / `ST_2` in 2019–2022, randomly samples a fixed number per year × site, and writes `R1.fasta`.
`HIV_subsample_count.py`	Reads `R1.fasta` and prints a year × site count table with row and column totals.
`Supplementary_Table_1.csv`	Supplementary Table 1 with the SUBSAMPLE column recomputed as `round(incidence × 21.81)`
`NAME_DATE_REPS.xlsx`	Name/date tables for the replicas, one sheet each (`R1` – `R5`), 32,182 sequences per sheet (the 1,896 subsampled CIENI-CEC sequences plus all background sequences). Tip-date file for the time-scaled phylogeny.

Pipeline

```
input.fasta ──► HIV_subsample_incidence.py ──► R1.fasta ──► HIV_subsample_count.py ──► count ta
                                                  │
                                                  └─► (combined with background sequences outs
                                                        ──► NAME_DATE_REPS.xlsx, sheet R1
```

Expected FASTA header format

Both scripts split the header on `|` and expect:

```
<ID>|<country>|<site>|<cluster>|<YYYY/MM/DD>
  0      1        2       3          4
```

For example, `SEQID|MX|ST_1|Cluster_NA|2019/07/17`. The site comes from field 2 and the year from the text before the first `/` in field 4.

Subsample targets

Hard-coded in `HIV_subsample_incidence.py`, derived from the lowest sequences-to-incidence ratio per year per location (DGE incidence; see Supplementary Table 1):

`ST_1` = CDMX, `ST_2` = MEX.

Year	Location	Available	Incidence †	Seqs/incidence	Sampled
2019	CDMX	571	14.2	40.21	310
2019	MEX	229	10.5	**21.81** ‡	229
2020	CDMX	705	7.6	92.76	166
2020	MEX	254	5.1	49.80	111
2021	CDMX	1931	13.2	146.29	288
2021	MEX	738	9.8	75.31	214
2022	CDMX	1899	14.4	131.88	314
2022	MEX	751	12.1	62.07	264
Total		**7078**			**1896**

† New HIV cases per 100,000 inhabitants per year, DGE (Greater Mexico City). ‡ Smallest sequences-to-incidence ratio; this cell sets the scale and is kept whole.

Derivation. Each target is `round(incidence × 21.81)`, where 21.81 is the smallest sequences-to-incidence ratio across the eight cells (MEX 2019 = 229 / 10.5). All eight hard-coded targets reproduce exactly under this rule, so the subsample is proportional to reported incidence rather than to how much each location was sequenced. "Available" is the full CIENI-CEC count for that year and location, and matches the supplementary table.

Requirements

- Python 3
- Biopython (`pip install biopython`)

Usage

```
python HIV_subsample_incidence.py   # input.fasta -> R1.fasta
python HIV_subsample_count.py       # prints the count table for R1.fasta
```